WORKED SOLUTION

STEP I 2018, Question 10

Mechanics • Connected Particles

STEP I 2018, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

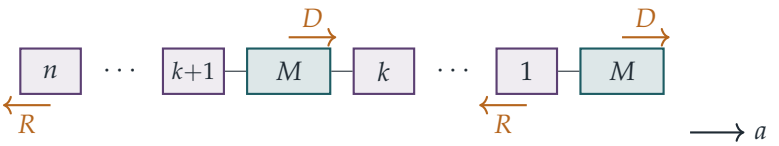
step.maths.org/questions/18-s1-q10

The key idea

Every vehicle has the same acceleration, so we find it once from the whole train and then apply Newton's second law to whichever piece of the train is most convenient: cutting the train at a coupling exposes that coupling's tension as an external force. The trick that makes the later parts easy is to work out the front tension T in two ways, from the front engine and from everything behind it. Doing so ties T to a single quantity, $R + ma$, which also governs every other coupling.

Solution

Model. Treat each engine and carriage as a particle moving along the horizontal track, and the couplings as light rods that can be in tension or compression. The engines feel no resistance, and the vertical forces on each vehicle balance, so only horizontal forces matter. Measure the acceleration a in the direction of motion. Number the carriages $1, 2, \ldots, n$ from the front, so that carriages $1, \ldots, k$ are between the engines and $k+1, \ldots, n$ are behind the second engine (with $1 \le k \le n-1$).



(Each carriage has a resistance R acting backwards; only two are drawn.)

The acceleration. The whole train has mass $nm + 2M$, and the couplings are internal, so

$$(nm + 2M)a = 2D - nR, \qquad a = \frac{2D - nR}{nm + 2M} > 0,$$

positive because $2D > nR$. In particular $R + ma > 0$, a fact we use repeatedly.

Part (i)

The front engine is pulled forward by D and backward by the tension T in its coupling, so

$$D - T = Ma.$$

Hence

$$\begin{aligned} T = D - \frac{M(2D - nR)}{nm + 2M} &= \frac{D(nm + 2M) - 2MD + nMR}{nm + 2M} \\ &= \frac{n(mD + MR)}{nm + 2M}, \end{aligned}$$

as required.

A neater form of T. Now look instead at everything behind the front engine: one engine and n carriages, of total mass $M + nm$. It is pulled forward by T and by the second engine's D, and resisted by nR:

$$T + D - nR = (M + nm)a.$$

Subtracting $D - T = Ma$ from this eliminates D:

$$2T - nR = nma, \qquad \text{so} \qquad T = \tfrac{1}{2}n\,(R + ma).$$

Part (ii)

Every coupling other than the front one is of one of two kinds.

Couplings in front of the second engine. Let T_j be the tension in the coupling just behind carriage j, for $1 \le j \le k$ (so T_k is the coupling between carriage k and the second engine). The front engine and carriages $1, \ldots, j$ have mass $M + jm$ and are pulled back by T_j:

$$D - jR - T_j = (M + jm)a \quad \Longrightarrow \quad T_j = (D - Ma) - j(R + ma) = T - j(R + ma).$$

Since $R + ma > 0$ and $j \ge 1$, we have $T_j < T$ for every such coupling, whatever k is.

Couplings behind the second engine. Let S_j be the tension in the coupling just in front of carriage j, for $k + 1 \le j \le n$. It is easiest to look at the vehicles behind it: carriages $j, \ldots, n$, of which there are $n - j + 1$, pulled forward only by S_j:

$$S_j - (n - j + 1)R = (n - j + 1)ma \quad \Longrightarrow \quad S_j = (n - j + 1)(R + ma).$$

? **Ask yourself** Which of these rear couplings carries the largest tension, and how does it compare with T?

The tension S_j decreases as j increases, so the largest is $S_{k+1} = (n-k)(R+ma)$, in the coupling directly behind the second engine. Comparing with $T = \frac{1}{2}n(R+ma)$, and dividing by $R+ma > 0$,

$$S_{k+1} < T \iff n-k < \tfrac{1}{2}n \iff k > \tfrac{1}{2}n.$$

So if $k > \frac{1}{2}n$, every rear coupling has tension at most $S_{k+1} < T$, and every front coupling has $T_j < T$. Therefore T is greater than the tension in any other coupling, as required. (The condition is also necessary: if $k \leq \frac{1}{2}n$ then $S_{k+1} \geq T$.)

Part (iii)

Take the coupling between carriage k and the second engine. From part (ii) and $T = \frac{1}{2}n(R+ma)$,

$$T_k = T - k(R+ma) = \left(\tfrac{1}{2}n - k\right)(R+ma).$$

If $k > \frac{1}{2}n$, the first factor is negative and the second is positive, so $T_k < 0$: this coupling is in compression. Physically, the second engine is pushing the k carriages in front of it, which is more than half the train, while the front engine pulls them.

Checks

- Take $n = 5$, $k = 3$, $M = 10$, $m = 2$, $D = 20$, $R = 1$. Then $a = \frac{35}{30} = \frac{7}{6}$ and the formula gives $T = \frac{5(40+10)}{30} = \frac{25}{3}$, matching $D - Ma = 20 - \frac{35}{3}$. Going back along the train the tensions are $\frac{25}{3}, 5, \frac{5}{3}, -\frac{5}{3}$ (carriage 3 to engine), then $\frac{20}{3}, \frac{10}{3}$. So T is the largest, and exactly one coupling is in compression, as predicted.
- If $M = 0$ the engines are just forces, and $T = D$, as it must be for a massless front engine.
- Each tension is a force times a dimensionless ratio, so the units are correct.

Common slips

- Giving the engines a resistance as well: the question puts resistance only on the carriages.
- Forgetting that the second engine's driving force acts on any section of train that contains it.
- Assuming every coupling must be in tension; a coupling is a rod, so a negative tension is a perfectly physical push.

Beyond STEP

Splitting a system into pieces and drawing a separate free-body diagram for each, with internal forces appearing in equal and opposite pairs, is the basis of the mechanics

of connected bodies. In first-year courses this becomes the analysis of trusses and frameworks, where you solve for internal forces in many members at once and some members turn out to be struts in compression rather than ties in tension. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats free-body diagrams and Newton's third law carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.